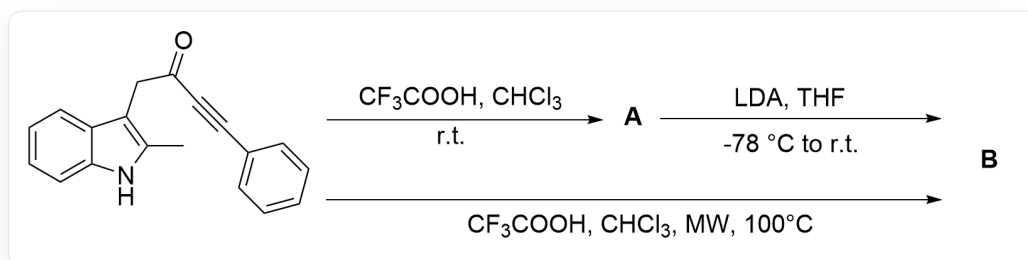


Question



This figure depicts a one-step organic reaction. The substrate is CC1=C(CC(C#CC2=CC=CC=C2)=O)C3=CC=CC=C3N1, which is converted to **A** under CF3COOH, CHCl3, r. t. conditions. **A** is converted to **B** under LDA, THF, -78°C to r. t. conditions; the substrate can be directly converted to **B** under CF3COH, CHCl3, MW, 100°C conditions.

Regarding the organic reaction shown in the figure above, it is known that:

- The ^1H – NMR information of **A** is:
 δ 7.66(d, 1H), 7.43 ~ 7.38(m, 1H), 7.31(t, 1H), 7.23 ~ 7.17(m, 4H), 7.00 ~ 6.96(m, 2H), 6.88(s, 1H), 2.83(d, 1H), 2.73(d, 1H), 2.21(s, 3H).
- B** contains a quinoline structure.
- The molecular weights of the substrate and **A, B** are all the same.

Which of the following statements is correct:

- A.** All other options are incorrect
- B.** **A** contains only one five-membered ring
- C.** **A** does not contain chiral carbon atoms.
- D.** The transformation of **A** to **B** involves a five-membered-ring intermediate.
- E.** **B** exhibits a bonding relationship: $\text{Ph} - \text{C} - \text{C} - \text{C} - \text{CH}_3$
- F.** **B** exhibits a bonding relationship: $\text{Ph} - \text{C} - \text{C} - \text{C} - \text{N}$

Answer

Correct Answer: **D**

Detailed Explanation

Observation of the ^1H NMR spectrum of **A**: Located in the low field, with a chemical shift greater than 6.5, containing 10 hydrogen atoms, it should be aromatic hydrogen; five of them belong to the phenyl group, and four belong to the benzene ring hydrogen of benzopyrrole. Therefore, it is speculated that **A** contains a phenyl benzylic $\text{sp}^2\text{C}-\text{H}$.

CHECKPOINT

1 PTS

A contains a phenyl benzylic $\text{sp}^2\text{C}-\text{H}$

The chemical shift of 2.21 corresponds to methyl hydrogen atoms, and the two d-peaks at 2.83 and 2.73 are clearly methylene peaks. Therefore, the methylene and methyl groups of the substrate did not change.

CHECKPOINT

1 PTS

The methylene and methyl groups of the substrate did not change

Considering the reactivity of the substrate, the nucleophilic site is the 3-position of pyrrole, and the electrophilic sites are the carbonyl group and the alkyne bond; the nucleophilic carbonyl group forms an unstable three-membered ring, so the 3-position of pyrrole undergoes nucleophilic attack on the alkyne bond, resulting in a 5-endo-dig reaction to form a five-membered spirocycle. The product structure is CC1=NC2=CC=CC=C2C13C(C4=CC=CC=C4)=CC(C3)=O, which is consistent with the analysis of the ^1H NMR spectrum, and therefore is structure **A**.

CHECKPOINT

1 PTS

The 3-position of pyrrole undergoes nucleophilic attack on the alkyne bond, resulting in a 5-endo-dig reaction to form a five-membered spirocycle

CHECKPOINT

1 PTS

Structure **A** is CC1=NC2=CC=CC=C2C13C(C4=CC=CC=C4)=CC(C3)=O

The transformation of **A** to **B** occurs in a basic environment. First, the methylene hydrogen at the α -position of the carbonyl group is abstracted to form an enolate anion, with the structure CC1=NC2=CC=CC=C2C13C(C4=CC=CC=C4)=CC([O-])=C3.

CHECKPOINT

1 PTS

Abstraction of the methylene hydrogen at the α -position of the carbonyl group to form an enolate anion

Then, the enolate anion nucleophilically attacks the electrophilic 2-position of pyrrole in the system, forming a five-fused-three-fused-five-membered ring intermediate, with the structure CC12[N-]C3=CC=CC=C3C14C2C(C=C4C5=CC=CC=C5)=O. Therefore, option D is correct.

CHECKPOINT

1 PTS

The enolate anion nucleophilically attacks the electrophilic 2-position of pyrrole in the system, forming a five-fused-three-fused-five-membered ring intermediate

CHECKPOINT

1 PTS

Five-fused-three-fused-five-membered ring intermediate, with the structure CC12[N-]C3=CC=CC=C3C14C2C(C=C4C5=CC=CC=C5)=O

Since **B** contains a quinoline structure, the three-membered ring opens to form a benzo six-membered ring, which isomerizes into a quinoline structure; therefore, the structure of **B** is CC1=NC2=CC=CC=C2C3=C1C(CC3C4=CC=CC=C4)=O.

CHECKPOINT

1 PTS

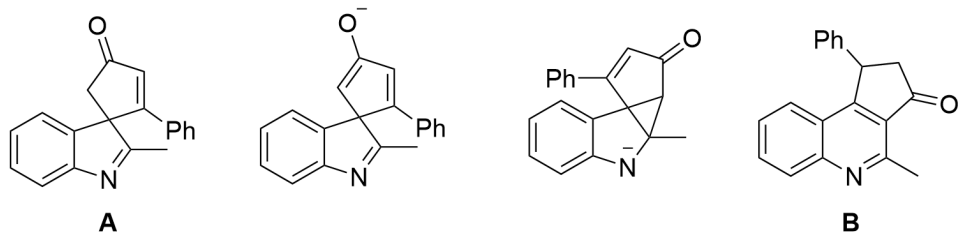
The three-membered ring opens to form a benzo six-membered ring, which isomerizes into a quinoline structure

CHECKPOINT

1 PTS

Structure **B** is CC1=NC2=CC=CC=C2C3=C1C(CC3C4=CC=CC=C4)=O

According to the structure, options B, C, E, and F are all incorrect.



A structure is CC1=NC2=CC=CC=C2C13C(C4=CC=CC=C4)=CC(C3)=O ; enolate anion, structure is CC1=NC2=CC=CC=C2C13C(C4=CC=CC=C4)=CC([O-])=C3 ; five-fused-three-fused-five-membered ring intermediate, structure is CC12[N-]C3=CC=CC=C3C14C2C(C=C4C5=CC=CC=C5)=O ; **B** structure is CC1=NC2=CC=CC=C2C3=C1C(CC3C4=CC=CC=C4)=O